

SPHERE: MISSION CONTROL

STUDENT LABORATORY GUIDE & EXPERIMENTAL TASK SHEET

Your Mission Challenge: Build a waterproof spacesuit capsule for a simulated astronaut (100 ml of hot water at 80°C), submerge it in an ice bath (0°C), and stream live temperature readings to your Mission Control dashboard. You'll compare your real experimental curve against a live computer model (Digital Twin) running Newton's Law of Cooling, and see how closely your hardware matches physics theory!

1. THE SPACE ENVIRONMENT & MISSION OBJECTIVES

Space is one of the most extreme environments humans have ever ventured into. In Low Earth Orbit, spacecraft and astronauts orbiting Earth cycle between violent temperature extremes every 90 minutes. In full sunlight, solar radiation bakes spacesuit exteriors up to $+121^{\circ}\text{C}$. When orbiting into Earth's shadow, ambient temperatures plunge to -157°C as heat radiates away into deep space.

Without an atmosphere to carry heat away gently, spacesuits rely on specialized **Multi-Layer Insulation (MLI)**. These blankets combine reflective foils that bounce radiant heat back with fluffy spacer materials that stop direct heat conduction.

In this lab, your crew will test different insulation setups to discover how well materials fight heat loss. By logging real sensor telemetry and comparing it with a theoretical cooling curve, you will calculate your crew's **Thermal Correlation Score**.

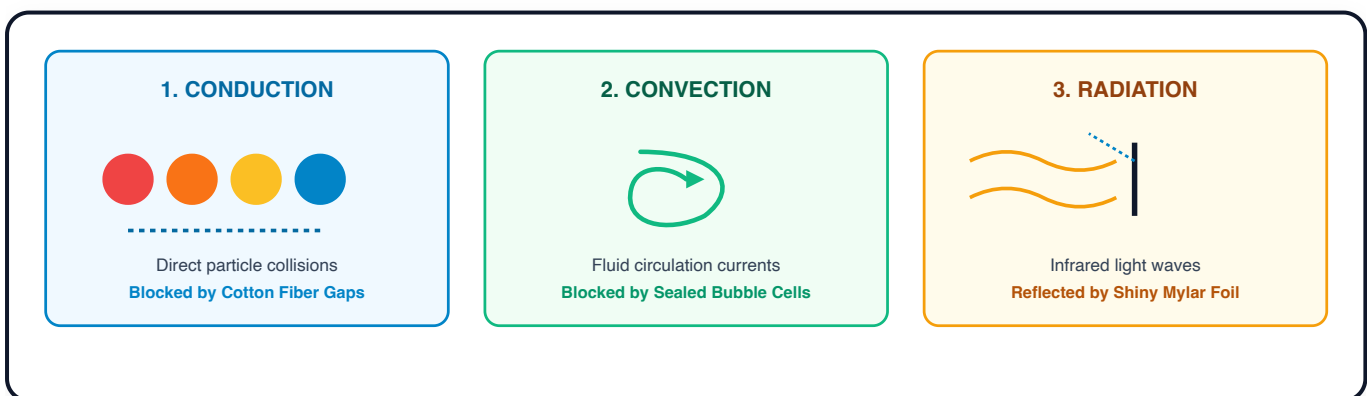


FIGURE 2: THREE MODES OF HEAT TRANSFER & THERMAL INSULATION BARRIERS

2. HARDWARE SETUP & SCHEMATIC

Your team will assemble a rugged, waterproof capsule designed to keep sensors dry and securely submerged:

HARDWARE PART	DESCRIPTION	WHAT IT DOES IN OUR LAB
Core Jar	100 mL wide-mouth PET jar	Holds our hot water core (80°C), representing the astronaut's internal body heat.
Cable Gland	PG7 waterproof nylon gland	Mounted through the jar cap to clamp the sensor wire and guarantee a 100% watertight seal.
Digital Sensor	Waterproof digital thermometer	Reads the internal water temperature with $\pm 0.5^{\circ}\text{C}$ accuracy and streams it live to your screen.
Ice Bath Chamber	Ice-water slurry (0°C)	Acts as our extreme cold space environment sink.

Quick Assembly Steps:

1. **Feed the Sensor:** Unscrew the dome cap on the PG7 gland and slide the metal temperature probe through the gland body.
2. **Mount on Cap:** Push the threaded gland through the hole in the jar cap, and tighten the locknut on the underside so it won't budge.
3. **Tighten the Seal:** Hand-tighten the outer dome nut until the rubber ring clamps snugly around the cable. This stops any water from leaking in.
4. **Center the Probe:** Make sure the metal sensor tip hangs right in the center of the jar without touching the bottom or sides.

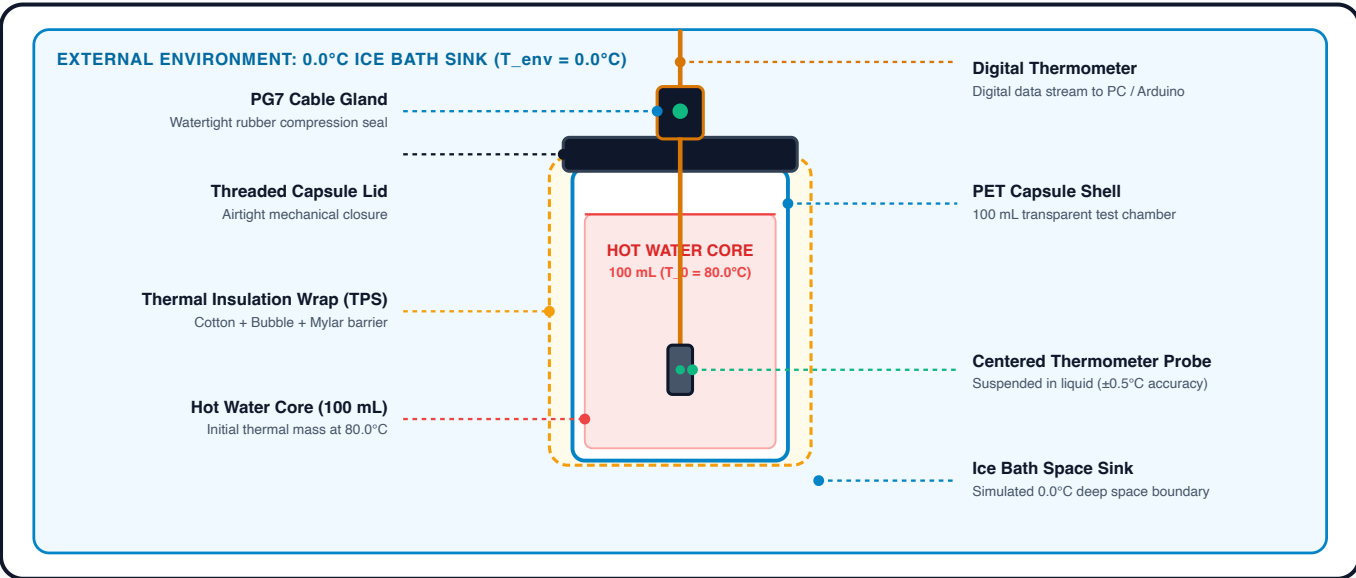


FIGURE 1: EXPERIMENTAL HARDWARE ASSEMBLY & SENSOR ALIGNMENT

3. THE 4 INSULATION CONFIGURATIONS

Each crew will test one of four insulation setups to see how different materials tackle heat loss:



FIGURE 3: MULTI-LAYER INSULATION (MLI) COMPOSITE LAYER STACK

4. THE PHYSICS: NEWTON'S LAW OF COOLING

Heat always flows from warmer objects to cooler surroundings. Over 300 years ago, Sir Isaac Newton discovered that the rate at which an object cools is proportional to the temperature difference between the object and its environment.

NEWTON'S LAW OF COOLING (EXPONENTIAL DECAY MODEL)

$$T(t) = T_{\text{env}} + (T_0 - T_{\text{env}}) \cdot e^{-k \cdot t}$$

Where:

$T(t)$ = Predicted core temperature (°C) at minute t

T_{env} = Ice bath temperature (0.0°C)

T_0 = Starting core temperature (80.0°C)

k = Cooling rate constant (min^{-1})

What does k mean? The cooling constant k measures how easily heat escapes. A *smaller* k value means superior insulation, a slower temperature drop, and a flatter cooling curve!

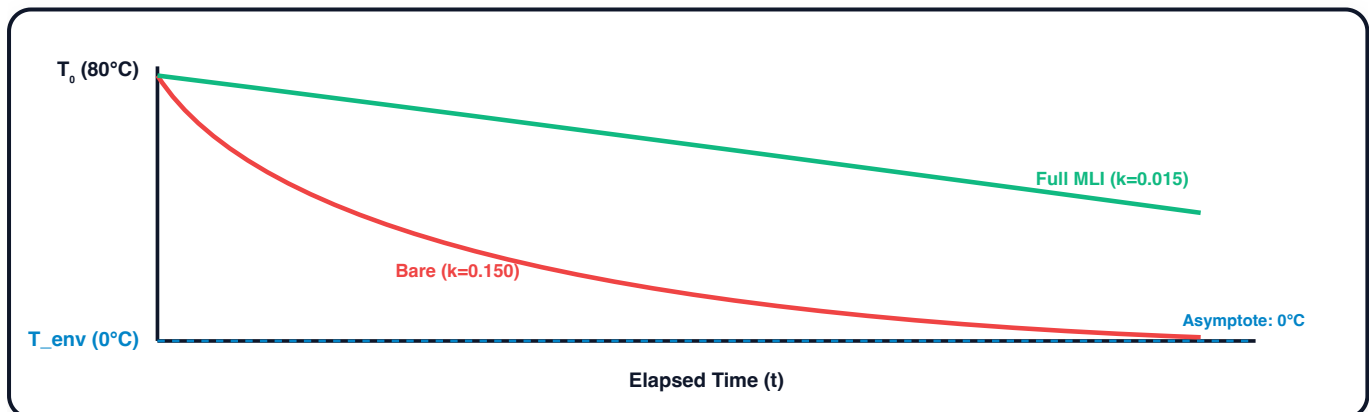


FIGURE 4: EXPONENTIAL DECAY PROFILES ACROSS HIGH VS LOW COOLING CONSTANTS (k)

5. STEP-BY-STEP LAB WORKFLOW

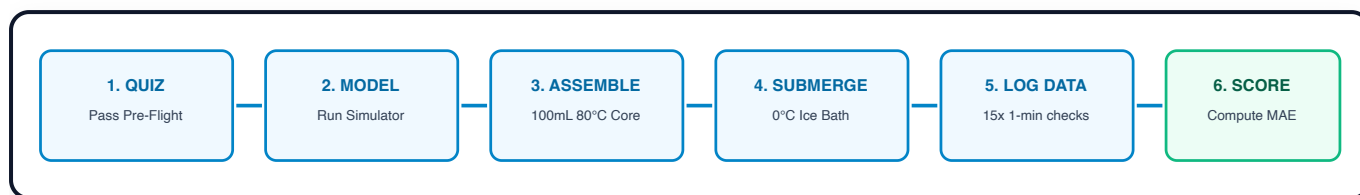


FIGURE 5: 6-STAGE EXPERIMENTAL PROTOCOL WORKFLOW

1. **Pass the Safety Quiz:** Open the Mission Control web app on your computer and complete the Pre-Flight Quiz to unlock live telemetry tools.
2. **Generate Model Prediction:** Go to the **Simulator** tab, choose your team's assigned insulation configuration, verify starting conditions ($T_0 = 80^\circ\text{C}$, $T_{\text{env}} = 0^\circ\text{C}$), and click **RUN SIMULATION PREDICTION**.
3. **Fill the Water Core:** Carefully measure 100 mL of hot water at 80°C , pour it into your jar, screw the cap on tightly, and verify the seal is leak-free.
4. **Submerge & Start Clock:** Lower the capsule into your 0°C ice bath and immediately click **START LIVE TIMER** in the **Live Telemetry** tab.
5. **Log Temperature Every Minute:** Every 60 seconds for 15 minutes, write down the observed temperature in your Task Sheet and enter it into Mission Control.
6. **Compare & Score:** Draw your real data points onto the graph grid, sketch your curve against the theoretical model line, and calculate your team's final Correlation Score!

FLIGHT CREW MEMBERS / NAMES:

MISSION CALLSIGN / TEAM ID:

DATE / CLASS PERIOD:

TESTED INSULATION
CONFIGURATION:

☐ A: Bare

☐ B: Cotton

☐ C: Mylar

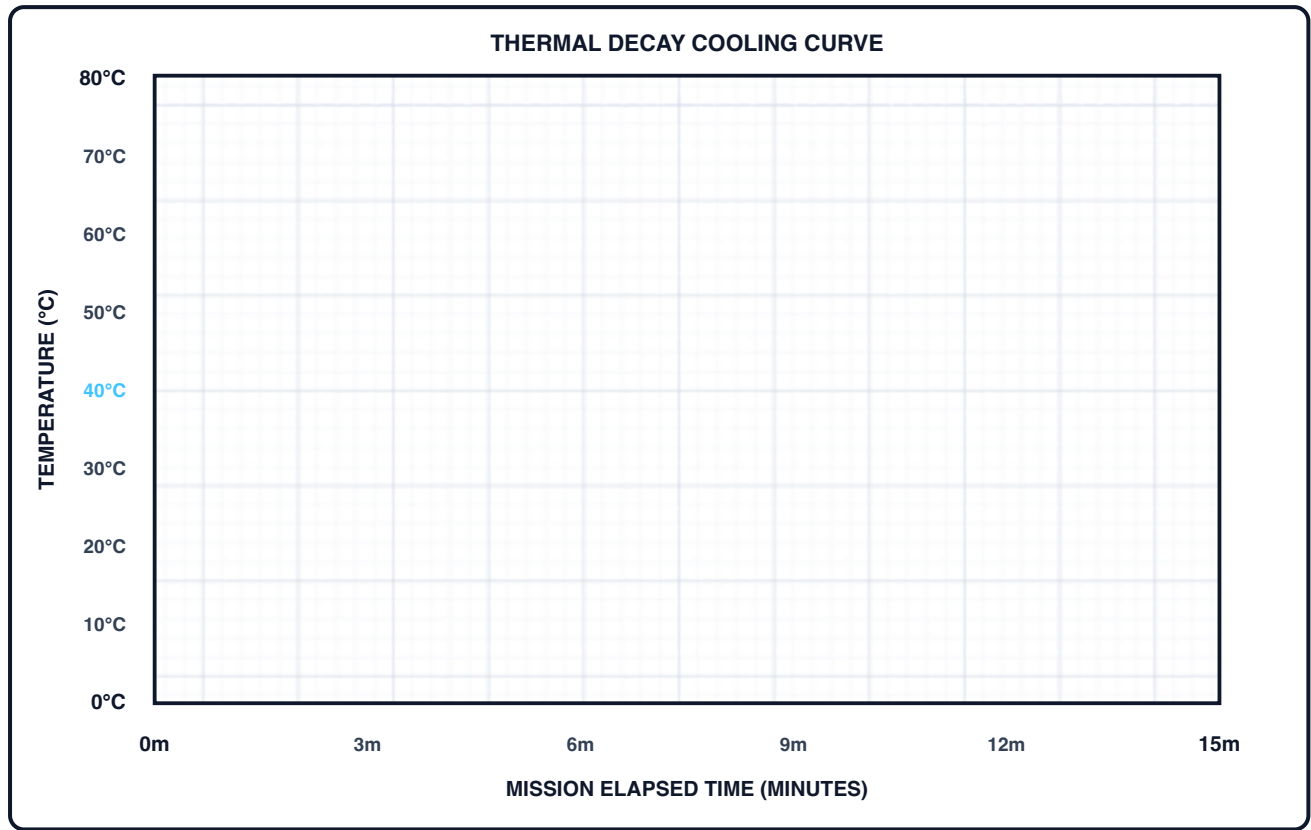
☐ D: Full MLI

Part A. Live Telemetry Data Logging Table

TIME (min)	MEASURED TEMP (°C)	PREDICTED TEMP (°C)	ABS DIFFERENCE $ \Delta T $ (°C)	OBSERVER NOTES
0 min	80.0°C	80.0°C	0.0°C	Capsule submerged in 0°C ice bath
1 min				Initial cooling rate check
2 min				
3 min				
4 min				
5 min				Mid-point check; check ice bath
6 min				
7 min				
8 min				
9 min				
10 min				Check curve flattening
11 min				
12 min				
13 min				
14 min				
15 min				Final measurement checkpoint

Part B. Coordinate Graph Grid

Plot your measured physical readings as solid dots (•) and connect them with a smooth line. Draw the theoretical model line as a dashed curve (---).



Part C. Thermal Correlation Score Calculation

Calculation Steps:

1. **Sum of Differences:** Add up all 15 numbers in the $|\Delta T|$ column: $\text{Sum} = \sum |\Delta T| =$ _____ °C
2. **Average Error:** Divide your sum by 15: $\text{Avg Error} = \text{MAE} = \frac{\text{Sum}}{15} =$ _____ °C
3. **Correlation Score:** Calculate your score: $\text{Score} = \max(0, 100\% - [\text{MAE} \times 5\%]) =$ _____ %

CREW MISSION RATING: ☐ **A+** ($\geq 95\%$) ☐ **A** (90–94%) ☐ **B** (80–89%) ☐ **C** (70–79%) ☐ **RE-TEST** ($< 70\%$)

7. LAB ANALYSIS & REFLECTION QUESTIONS

Question 1: Why doesn't shiny Mylar foil work by itself in cold water?

Mylar is famous for reflecting heat radiation in space. Why did Mylar alone perform poorly when dipped directly into the ice water bath (0°C) without cotton or bubble wrap underneath?

Question 2: What does the cooling constant (k) tell us?

In Newton's Law $T(t) = T_{\text{env}} + (T_0 - T_{\text{env}}) \cdot e^{-kt}$, does a *smaller* k value mean a better insulator or a worse insulator? Explain using the formula.

Question 3: How does Multi-Layer Insulation (MLI) work together?

Explain how combining Cotton + Bubble Wrap + Mylar in Configuration D creates a team effect that works far better than any single layer alone.

Question 4: Where could experimental errors come from?

List two physical reasons why your real sensor readings might differ from the smooth curve predicted by the computer model.

Question 5: What happens if your ice bath warms up?

If your classroom ice bath warms from 0.0°C to 4.0°C while testing, what should you change in the Mission Control simulator so the prediction stays accurate?

Question 6: What if we double the water volume?

If we doubled the water inside the jar from 100 ml to 200 ml, would it cool down faster or slower? Would k increase or decrease?

Question 7: Spacesuit Design Trade-Offs

If adding insulation layers keeps astronauts warmer, why don't space engineers just wrap a spacesuit in 50 layers of MLI? What problems would that cause for an astronaut on a spacewalk?

Question 8: How can you improve your Correlation Score?

Describe two practical things your crew could do to improve your Correlation Score from 75% to over 95% on a second run.